

Analysis note: position of the $Z \rightarrow \mu^+\mu^-$ resonance peak in the CMS 2012 **DoubleMuParked** Open Data skim

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Abstract

The dimuon invariant-mass spectrum is reconstructed from a skim of the CMS 2012 **DoubleMuParked** Open Data sample. Events with exactly two opposite-charge muons with $p_T > 20$ GeV and $|\eta| < 2.4$ are selected, and the Z resonance peak position is extracted with a binned χ^2 fit of a Voigtian signal plus an exponential background in the 85–97 GeV mass range. The fitted peak position is $m_{\mu\mu}^{\text{peak}} = 91.002 \pm 0.041$ (stat.) ± 0.122 (fit range) GeV; the difference with respect to the PDG world-average Z mass is -186 MeV (2.0 per mille). No muon momentum-scale or final-state-radiation corrections are applied, so the result is a peak-position determination, not a Z mass measurement.

1 Introduction

The Z boson decaying to two muons produces one of the cleanest signatures at a hadron collider: a narrow resonance at $m_Z = 91.1880 \pm 0.0020$ GeV [1] on top of a small, smoothly falling Drell–Yan and misreconstruction background. This note documents a self-contained determination of the dimuon peak position in real proton–proton collision data recorded by the CMS detector [2] in 2012 at $\sqrt{s} = 8$ TeV and released through the CERN Open Data portal [3]. The goal is a reproducible analysis bundle: every number quoted here is generated by `analysis.py` and imported into this document as a macro, so the note and the code cannot drift apart.

2 Data

The input is `zmumu_skim.root`, a skim of the CMS 2012 **DoubleMuParked** primary dataset published on the CERN Open Data portal (<https://opendata.cern.ch>) and converted to a flat NanoAOD-like format with the CMS `AOD2NanoAODOutreachTool` [3]. The skim contains 83343 events with jagged per-muon branches `Muon_pt`, `Muon_eta`, `Muon_phi`, `Muon_mass`, and `Muon_charge` (momenta and masses in GeV). Only raw event counts are used throughout: no luminosity normalization, pile-up reweighting, or trigger-efficiency correction is applied, none of which affect the position of the resonance peak.

3 Event selection

Muons are required to have $p_T > 20$ GeV and $|\eta| < 2.4$, matching the muon-system acceptance [2]. Events must contain exactly two such muons with opposite charge. The dimuon invariant mass is computed from the four-momentum sum of the two muons, and the spectrum is analyzed in the $60 \leq m_{\mu\mu} < 120$ GeV window. The cutflow is given in Table 1.

Table 1: Event selection cutflow. Raw (unweighted) event counts.

Selection	Events
All events in skim	83343
Exactly two muons, $p_T > 20$ GeV, $ \eta < 2.4$	11983
Opposite charge	10420
$60 \leq m_{\mu\mu} < 120$ GeV	8589

4 Method

The selected masses are histogrammed in 0.5 GeV bins. The spectrum is fitted by minimizing a χ^2 with Poisson bin uncertainties, using the model

$$\frac{dN}{dm} = n_{\text{sig}} V(m - \mu; \sigma, \Gamma_Z/2) + n_{\text{bkg}} f_{\text{bkg}}(m), \quad (1)$$

where V is a Voigtian — a relativistic-width Breit–Wigner convolved with a Gaussian resolution of width σ — and f_{bkg} is a falling exponential normalized over the fit range, so that n_{sig} and n_{bkg} are yields in that range. The Breit–Wigner width is fixed to the PDG Z width $\Gamma_Z = 2.4955$ GeV [1]; the Gaussian σ absorbs the detector resolution. The nominal fit uses the 85–97 GeV core of the peak, excluding the low-mass tail populated by final-state radiation (FSR), which the symmetric Voigtian does not model.

The sensitivity of the peak position to this choice is assessed by repeating the fit with three alternative ranges (80–100, 83–99, and 86–96 GeV, as defined in `analysis.py`); the largest absolute shift of the fitted peak is taken as a fit-range systematic uncertainty.

5 Results

The nominal fit converges with $\chi^2/\text{ndf} = 25.1/19 = 1.32$ and yields

$$m_{\mu\mu}^{\text{peak}} = 91.002 \pm 0.041 (\text{stat.}) \pm 0.122 (\text{fit range}) \text{ GeV}, \quad (2)$$

with a Gaussian resolution $\sigma = 1.19 \pm 0.07$ GeV and fitted yields of 6446 ± 191 signal and 978 background events in the fit range. The spectrum and fit are shown in Figure 1.

Compared with the PDG world average $m_Z = 91.1880 \pm 0.0020$ GeV [1], the difference $m_{\mu\mu}^{\text{peak}} - m_Z$ is -186 MeV, a relative difference of 2.0 per mille. This difference is comparable to the combined quoted uncertainties, and its sign is the one expected from the two dominant effects deliberately left uncorrected here: unrecovered FSR photons, which shift the reconstructed peak below the pole, and the muon momentum scale, which is not calibrated in this open-data skim.

Uncertainties not evaluated. Only the statistical uncertainty from the fit and the fit-range variation are quantified. Not evaluated are: the muon momentum-scale and resolution calibration (the dominant systematic in a real Z mass measurement), the modeling of FSR, the choice of signal parameterization (a symmetric Voigtian, with Γ_Z fixed) and of the background shape, and any alignment- or charge-dependent momentum biases. For this reason the result is presented as a determination of the reconstructed peak position, and no claim of a competitive Z mass measurement is made.

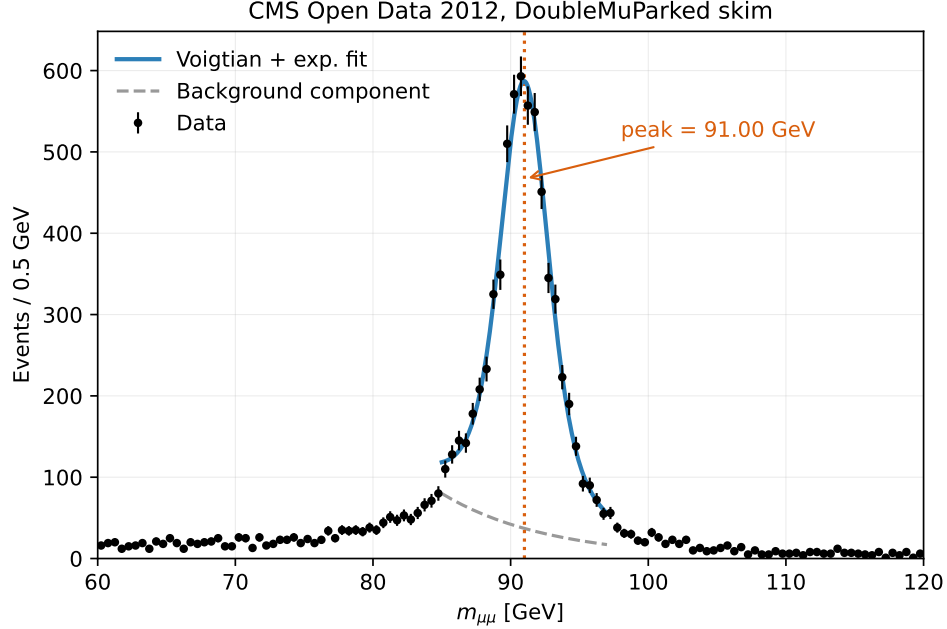


Figure 1: Dimuon invariant-mass spectrum after the full selection (points), with the Voigtian-plus-exponential fit in the 85–97 GeV range (solid line) and its background component (dashed line). The dotted vertical line marks the fitted peak position.

6 Conclusion

The $Z \rightarrow \mu^+ \mu^-$ resonance is clearly reconstructed in the CMS 2012 DoubleMuParked Open Data skim, with 8589 selected events in the 60–120 GeV window. The fitted peak position of 91.002 ± 0.041 (stat.) ± 0.122 (fit range) GeV agrees with the PDG Z mass at the per-mille level, demonstrating that the public skim, the selection, and the fitting chain are sound. The full analysis is reproducible from `analysis.py` as described in the bundle `README`.

References

- [1] S. Navas et al. Review of particle physics. *Phys. Rev. D*, 110(3):030001, 2024.
- [2] S. Chatrchyan et al. The CMS Experiment at the CERN LHC. *JINST*, 3:S08004, 2008.
- [3] Kati Lassila-Perini, Clemens Lange, Edgar Carrera Jarrin, and Matthew Bellis. Using CMS Open Data in research – challenges and directions. *EPJ Web Conf.*, 251:01004, 2021.